

unmodified server
(Redis · Memcached · Nginx · Node · PostgreSQL)

POSIX syscalls: sockets · time · randomness · threads

OneBarrier determinism shim (condition L)

LD_PRELOAD · no kernel · no app change

virtual clock
(time)

rng trap
(getrand/urnd)

share-nothing
shard

replay + output
suppress

durable ordered log + timestamp- T snapshot

1Pipe fabric:
global delivery order (O) +
reliable commit barrier (R)

in-fabric replicas
(durability, D)

uncoordinated recovery cut

1-RTT copy, inside the barrier